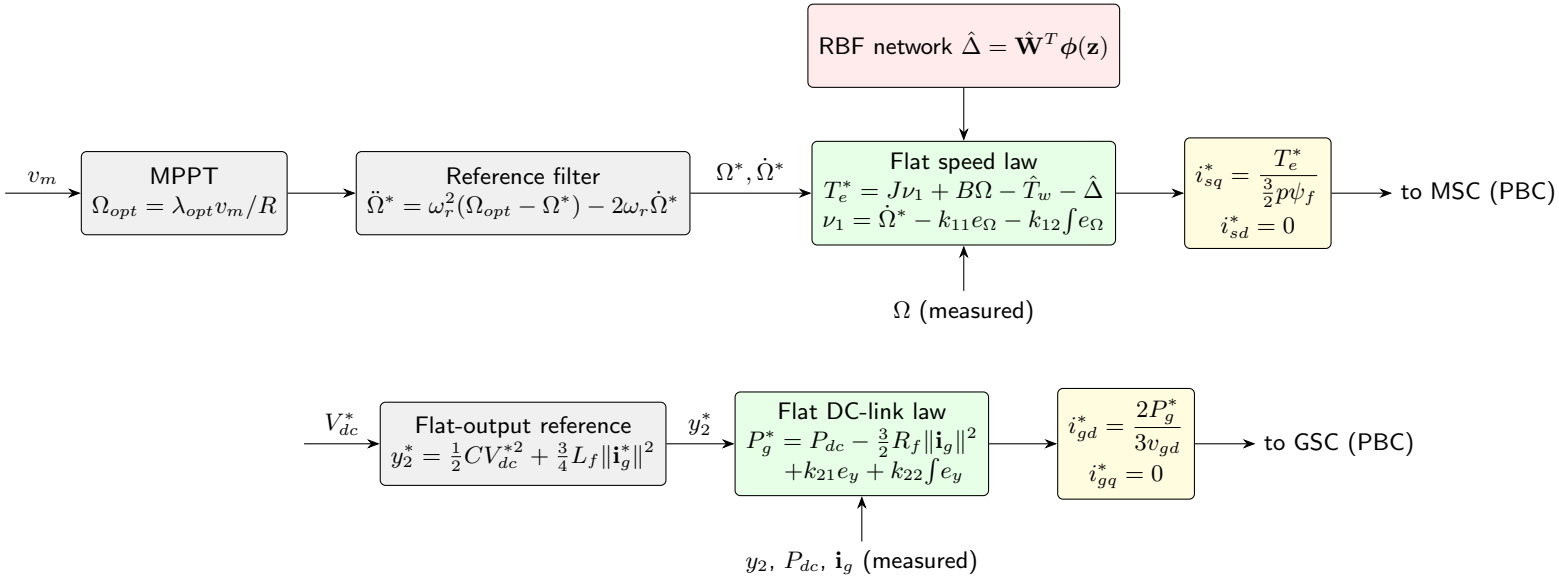

RBF network $\hat{\Delta} = \hat{\mathbf{W}}^T \boldsymbol{\phi}(\mathbf{z})$
v_m
MPPT
$\Omega_{opt} = \lambda_{opt} v_m / R$
Reference filter
$\ddot{\Omega}^* = \omega_r^2(\Omega_{opt} - \Omega^*) - 2\omega_r \dot{\Omega}^*$
$\Omega^*, \dot{\Omega}^*$
Flat speed law
$T_e^* = J\nu_1 + B\Omega - \hat{T}_w - \hat{\Delta}$
$\nu_1 = \dot{\Omega}^* - k_{11} e_\Omega - k_{12} \int e_\Omega$
$i_{sq}^* = \dfrac{T_e^*}{\frac{3}{2} p \psi_f}$
$i_{sd}^* = 0$
to MSC (PBC)
Ω (measured)
V_{dc}^*
Flat-output reference
$y_2^* = \frac{1}{2} C V_{dc}^{*2} + \frac{3}{4} L_f \|\mathbf{i}_g^*\|^2$
y_2^*
Flat DC-link law
$P_g^* = P_{dc} - \frac{3}{2} R_f \|\mathbf{i}_g\|^2$
$+ k_{21} e_y + k_{22} \int e_y$
$i_{gd}^* = \dfrac{2P_g^*}{3 v_{gd}}$
$i_{gq}^* = 0$
to GSC (PBC)
y_2, P_{dc}, $\mathbf{i}_g$ (measured)